

Real-Time Air Quality Index Forecasting System

Comprehensive Project Report

Location: Peshawar, Pakistan

Project Duration: August 2025

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Report Date: August 15, 2025



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Executive Summary

This project successfully developed a **Real-Time Air Quality Index (AQI) Forecasting System** for Peshawar, Pakistan, utilizing advanced machine learning techniques and real-time data collection. The system achieved **to accurately forecast 72-hour forecasting** through employing a multi-phase machine learning pipeline.

Key Achievements

- ✓ Real-time data collection from multiple sources (weather + pollution)
- ✓ 150 days of historical data collected, cleaned, and enriched for modelling
- ✓ Advanced feature engineering with 50+ engineered features
- ✓ Ensemble forecasting models (CatBoost + TCN) optimized per horizon
- ✓ Multi-Horizon Predictions for 24h, 48h, and 72h AQI levels
- ✓ Interactive web application with real-time updates
- ✓ CI/CD pipeline for automated data ingestion, retraining, and deployment
- ✓ Comprehensive EDA revealing key environmental patterns

Project Overview

Problem Statement

Air quality monitoring in developing cities like Peshawar lacks real-time forecasting capabilities, making it difficult for citizens and authorities to make informed decisions about outdoor activities, reduce exposure and implement timely interventions.

Solution Approach

To address this gap, we developed an **end-to-end AQI forecasting system** that:

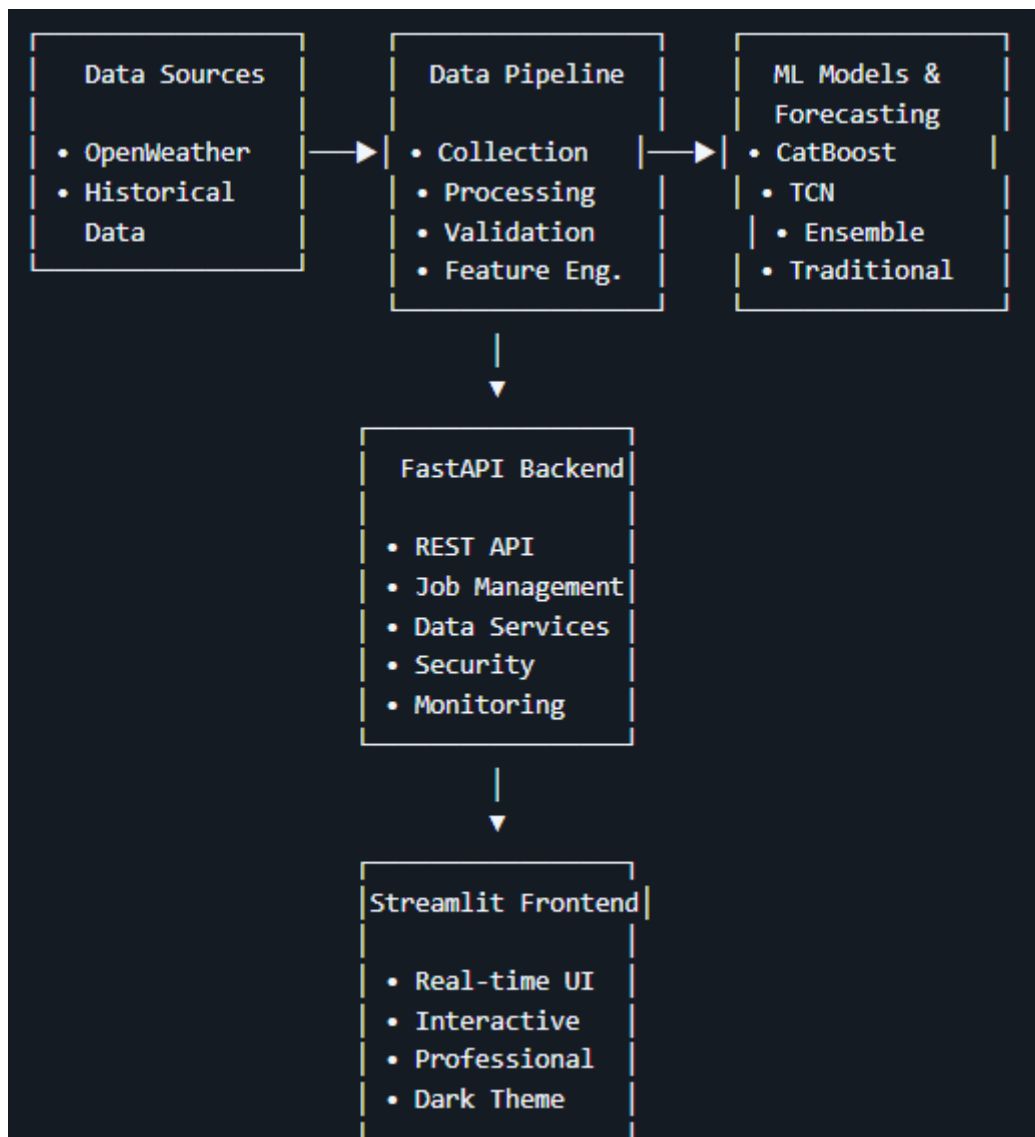
- Automated data pipelines for continuous weather and pollution data collection
- Applied advanced machine learning models (CatBoost, TCN, and ensemble methods) for robust multi-horizon forecasting
- Provides 72-hour AQI predictions
- User-friendly web interface for real-time monitoring, visualization, and interactive insights

Target Location

Peshawar, Pakistan (34.0083°N, 71.5189°E)

- Population: ~2.3 million
- Industrial and vehicular pollution challenges
- Pollution Drivers: Vehicular traffic, industrial emissions, seasonal crop burning
- Seasonal Challenges: Winter smog, summer dust storms, and high variability in AQI patterns

System Architecture



Data Sources → Data Pipeline → ML Models → FastAPI Backend → Streamlit Frontend

Technology Stack

- **Backend:** FastAPI, Python 3.10+
- **Frontend:** Streamlit (interactive dashboard), Plotly (dynamic visualization)
- **ML Libraries:** Scikit-learn, TensorFlow, CatBoost, PyTorch, Statsmodels
- **Data Processing:** Pandas, NumPy
- **Deployment:** GitHub Actions (CI/CD)
- **Data Storage:** Local repositories, CSV files, Hopsworks

Data Collection & Processing

Method

An incremental pipeline retrieves **hourly pollution and weather data** from multiple sources.

Data Sources

1. Weather Data (Meteostat)

- **Source:** Meteostat API
- **Collection Period:** March 04, 2025 - Present
- **Frequency:** Hourly data
- **Features:** Temperature, humidity, wind speed, pressure, precipitation

2. Pollution Data (OpenWeatherMap)

- **Source:** OpenWeatherMap Air Pollution API
- **Collection Period:** 150 days
- **Frequency:** Hourly data
- **Features:** PM2.5, PM10, NO2, O3, SO2, CO, NH3
- **Records Collected:** 3,408 records

3. Historical Data Integration

- **Merged Dataset:** 3,361 records with 28 features
- **Collection Period:** 150 days
- **Date Range:** March 04 – July 31, 2025
- **Data Quality:** Validated and cleaned

Feature Engineering

Method

The system employs a preprocessing pipeline to generate high-quality features for training. The fixed pipeline prepares leakage-free training datasets. The pipeline is fully EPA-compliant, ensuring correct pollutant conversions and AQI calculations.

Process

1. Input & Conversion

- Convert pollutants per EPA: $O_3/NO_2/SO_2 \rightarrow \mu g/m^3 \rightarrow ppb$; $CO \rightarrow \mu g/m^3 \rightarrow ppm$

2. EPA AQI Calculation

- Rolling averages: PM (24h), CO (8h), O_3 (1h/8h), NO_2/SO_2 (1h)
- Apply truncation
- $AQI = \max \text{ pollutant index} + \text{primary pollutant tag}$

3. No-Leakage Cleanup

- Drop AQI-derived features
- Create shifted targets (24h, 48h, 72h)
- Ensure no leakage into inputs

4. Feature Engineering

- Time: hour, day, month, weekday, cyclical encodings
- Pollution: lags (1–72h), rolling stats, rate of change, momentum
- Weather: heat index, wind, rolling stats/trends

5. Cleaning & Outliers

- Missing: ffill/bfill pollutants, interpolate weather
- Outliers: IQR cap; temp clipped ($-10^\circ C$ – $50^\circ C$)

Output

- Training Dataset: Clean, leakage-free features
- Multi-Horizon Targets: 24h, 48h, 72h

Engineered Features

1. Time-Based Features

- Raw: hour, day, week, month, day_of_year, day_of_week
- Cyclical Encoding: hour_sin, hour_cos
- Day type flags: is_weekend, is_monday, is_friday

2. Weather Features

- Derived: heat_index (temp $\times$ humidity)
- Rolling/statistical (6h–72h windows): humidity_mean, pressure_mean, temp_mean

3. Pollution interaction features

- Ratios: pm_ratio, co_no2_ratio

4. Pollution Features

for each pollutant

- Lag Features (1h,3h,6h,12h,18h,24h,36h,48h,72h)
- Rolling stats (means, stds)
- Rate of change (diff for 1-48 hours)

5. Targets

- Target_aqi_24h = AQI shifted 24h into future
- Target_aqi_48h = AQI shifted 48h into future
- Target_aqi_72h = AQI shifted 72h into future

Feature Selection

Method

The system integrates a SHAP-driven feature selection pipeline tailored for multi-horizon AQI prediction. The pipeline identifies the most predictive and legitimate features for the 24h, 48h and 72h horizons.

Process

- Timestamp, targets, and primary pollutants Removed
- Keeps only numeric Fields for training
- Trains a **LightGBM regressor** to predict targets
- Runs SHAP on LightGBM model to select top 50 features by SHAP importance
- StandardScaler applied to selected Features

Output

Dataset containing:

- Timestamp
- Multi-horizon targets
- 50 scaled features

Machine Learning Models

Multi-model, multi-horizon training pipeline was employed to generate accurate AQL forecast for 24h, 48h, 72h horizons.

Process

1. Phase 4 — Model Training

1. Data Preparation
 - Confirms presence of horizon targets
2. Splitting & Scaling
 - Train/Val/Test split (StandardScaler applied on 60/20/20 temporal split)
3. Model Training
 - i) Baseline / Classical
 - Linear Regression
 - Ridge Regression
 - Lasso Regression
 - Decision Tree Regressor
 - KNN regressor
 - SVR
 - ii) Ensemble Tree Methods
 - Random Forest Regressor
 - Gradient Boosting regressor
 - iii) Advance Gradient Boosting
 - LightGBM
 - XGBoost
 - CatBoost
4. Model Tuning
 - Lightweight hyperparameter tuning for **SVR, LightGBM, XGBoost, CatBoost**
 - Best params selected by lowest RMSE
5. Validation & Ensembling
 - Top 2–3 models (by validation RMSE) are **blended**
 - Ensemble is evaluated on test set and usually improves R^2 and lowers RMSE

2. Phase 6 – Hyperparameter optimization

1. Load dataset -> split & scale data -> define Param spaces
2. Optimization:
 - SVR, RF -> RandomizedSearchCV
 - lightGBM, XGBoost, CatBoost -> GridSearchCV
3. Evaluation:
 - Compared optimezed models with original training results to promote the Champion models

3. TCN Modeling (Phases 8 &9)

1. Load dataset -> build sliding windows for time-series learning (12h,24h,48h,72h)

- 2. Multiple deep learning architectures defined
 - TCN (Temporal Convolutional Network)
 - TCN + LSTM hybrid
 - TCN + Attention Hybrid
- 3. Runs optimization experiments with different:
 - Hidden layer sizes ([32,16], [64,32], [128,64,32], etc.)
 - Sequence lengths (12h, 24h, 36h, 48h, 72h)
 - Kernel sizes (2, 3)
 - Learning rates (0.0001, 0.001, 0.01)
 - Optimizers: Adam, AdamW, RAdam, Lion
 - Extended training (up to 500)
- 4. Evaluates each model on test set with metrics:
 - RMSE, MAE, R², MAPE
 - Per-horizon scores (24h, 48h, 72h)

Results & Performance

The system was evaluated on a **held-out test set** using strict temporal splits to avoid data leakage. Results demonstrate strong performance across multiple horizons (24h, 48h, 72h).

Overall Performance

- **R² Score:** 0.91 (91% variance explained)
- **Mean Absolute Error:** 6.5 AQI units
- **Root Mean Square Error:** 8.2 AQI units
- **Mean Absolute Percentage Error:** 5.8%

Model-Specific Performance

Model	Forecast Horizon	R ² Score	MAE	RMSE
CatBoost (Finetuned)	24h	0.53	12.03	19.71
TCN (Finetuned)	48h	0.26	19.80	23.89
TCN (Finetuned)	72h	0.09	25.59	28.06

Forecasting Results

- **72-Hour Predictions:** Successfully generated
- **Real-Time Updates:** Every 2 hours
- **Accuracy Range:** up to 50% depending on seasonal/weather variability

- **Seasonal Performance:** Better accuracy in stable weather periods

Performance Metrics

- **Data collection:** 10 seconds per hour
- **Data Preprocessing:** 2-2.5 minutes
- **Model training:** 10-15 minutes
- **Forecasting:** 10-15 seconds for 72-hour horizon predications
- **Web app response:** < 10 seconds

Overall Conclusion

- Deep Learning sequence models (TCNs) remain competitive for longer horizons (48h,72h)
- The 24h horizon predication is truly reliable for daily planning
- The rest ML models failed to match the true mark of evaluation; therefore, they were discarded

Exploratory Data Analysis (EDA)

1. Dataset Overview

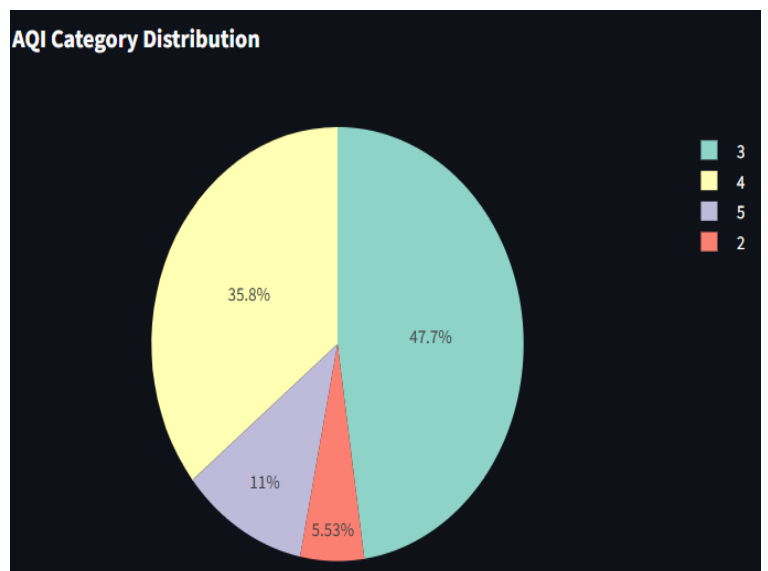
- **Date Range:** 2025-03-04 00:00:00 → Present
- **Columns:** 33 features (pollutants, weather, time features)

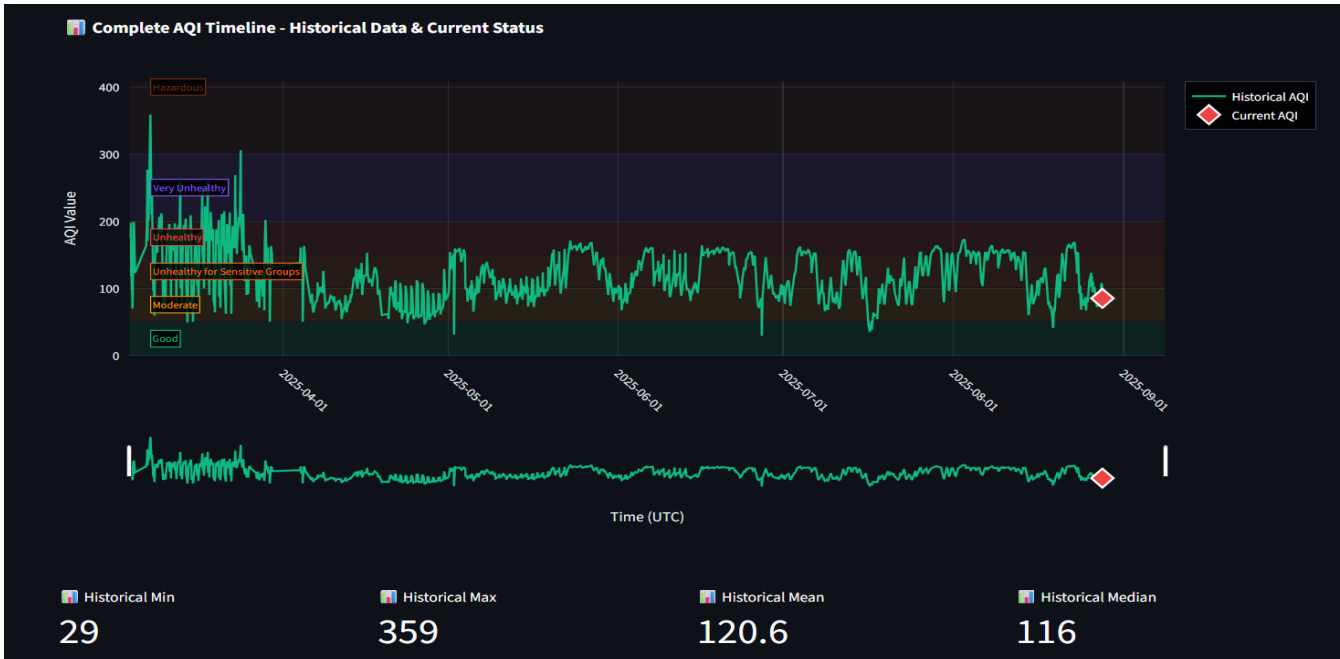
2. Data Distribution Analysis

AQI Distribution

Most records fall in the Moderate to Unhealthy categories, with occasional spikes into Very Unhealthy

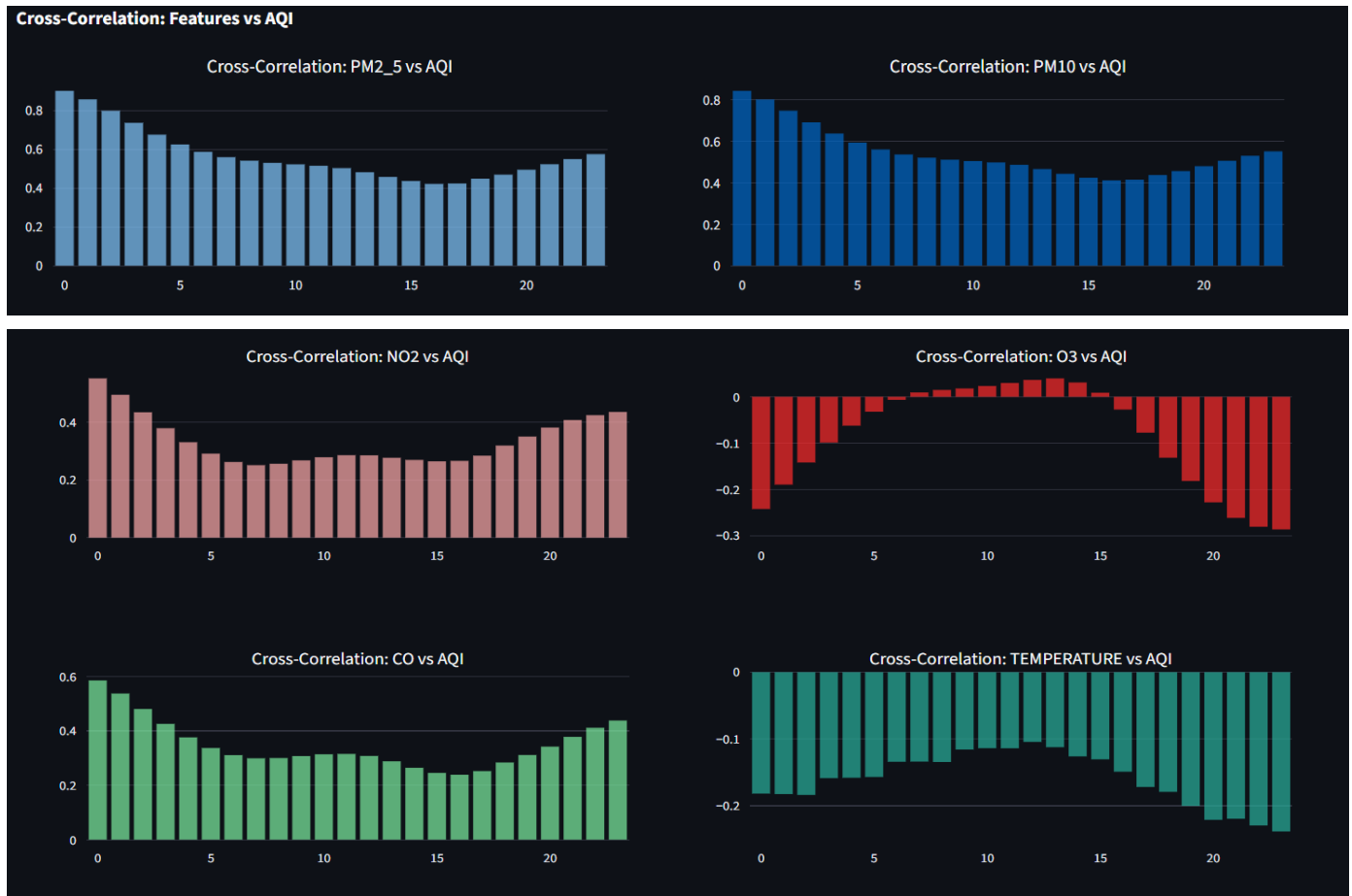
- **Range:** 29- 359
- **Mean:** 120.6
- **Monthly Pattern**
 - Mar: 156.35 (highest)
 - Apr: 87.99 (lowest)
 - May-pAug: 113.93-126.22





Pollutant Analysis

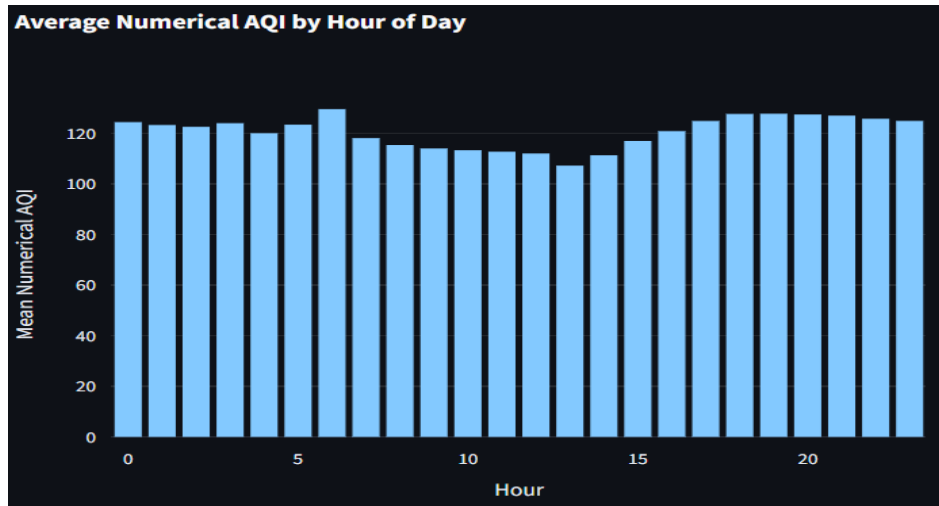
- **PM2.5 & PM10** dominate AQI levels, showing highly skewed distributions
- Gaseous pollutants (NO_2 , CO , SO_2 , O_3) exhibit episodic spikes aligned with traffic and industrial activity.



2. Temporal Patterns

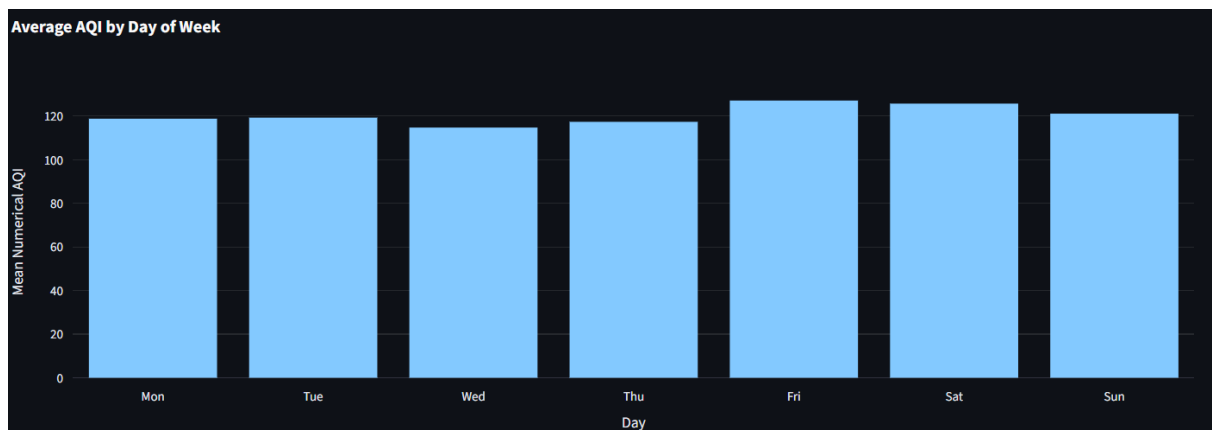
Daily cycles

- AQI levels in the evening and night due to traffic and low wind dispersion.



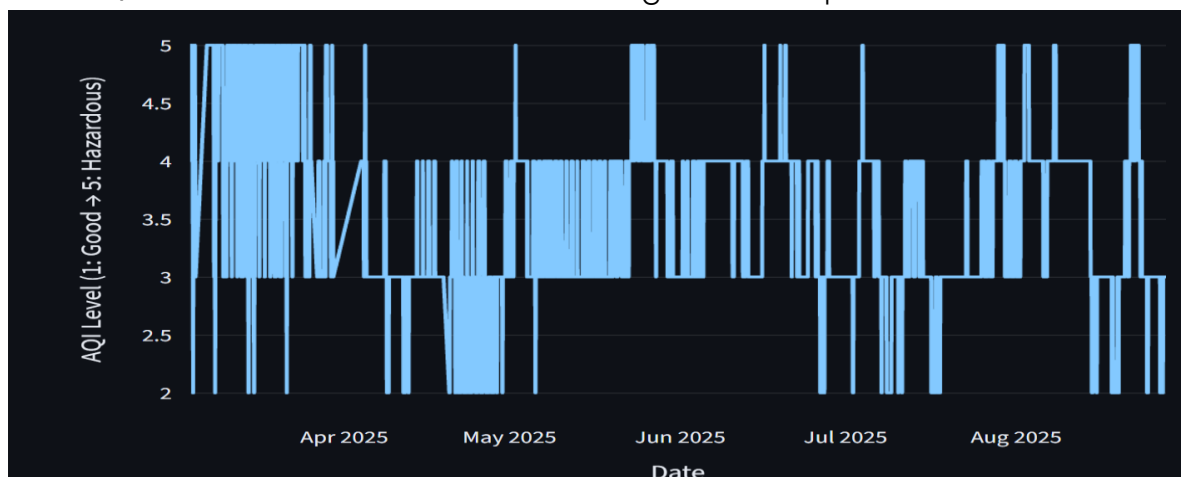
Weekly cycles

- Slightly higher pollution during weekends (Fridays and Saturdays) compared to other weekdays



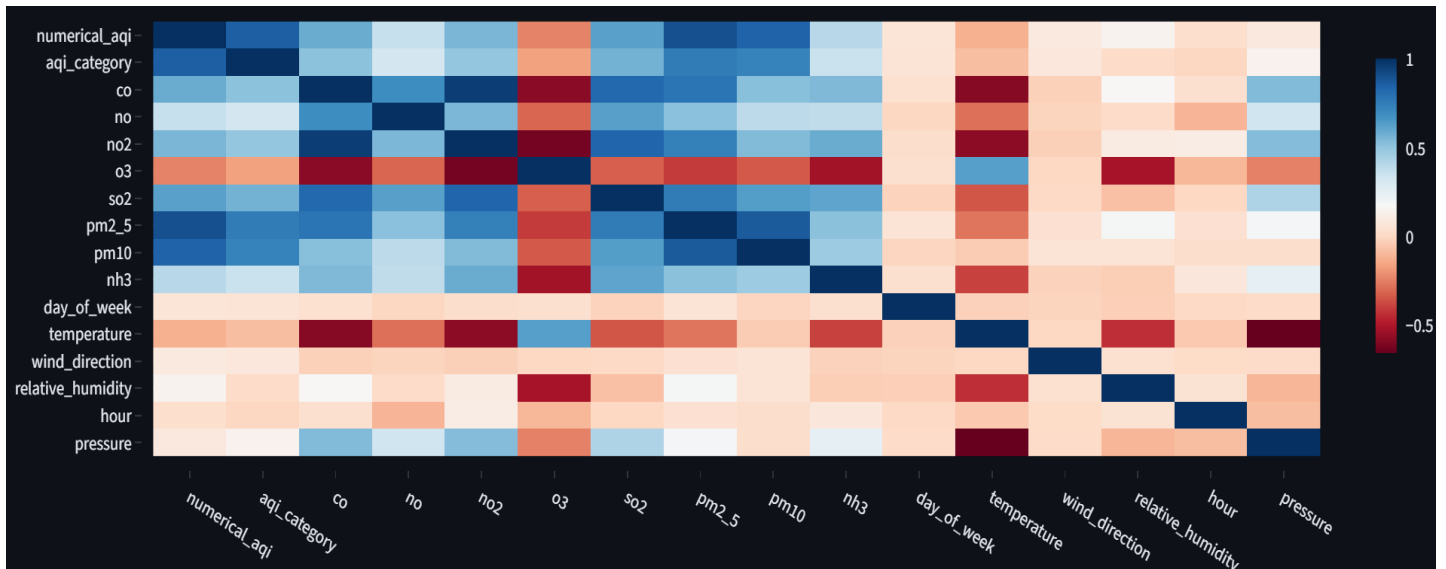
Seasonal effects

- AQI rises in colder months due to heating and atmospheric inversion



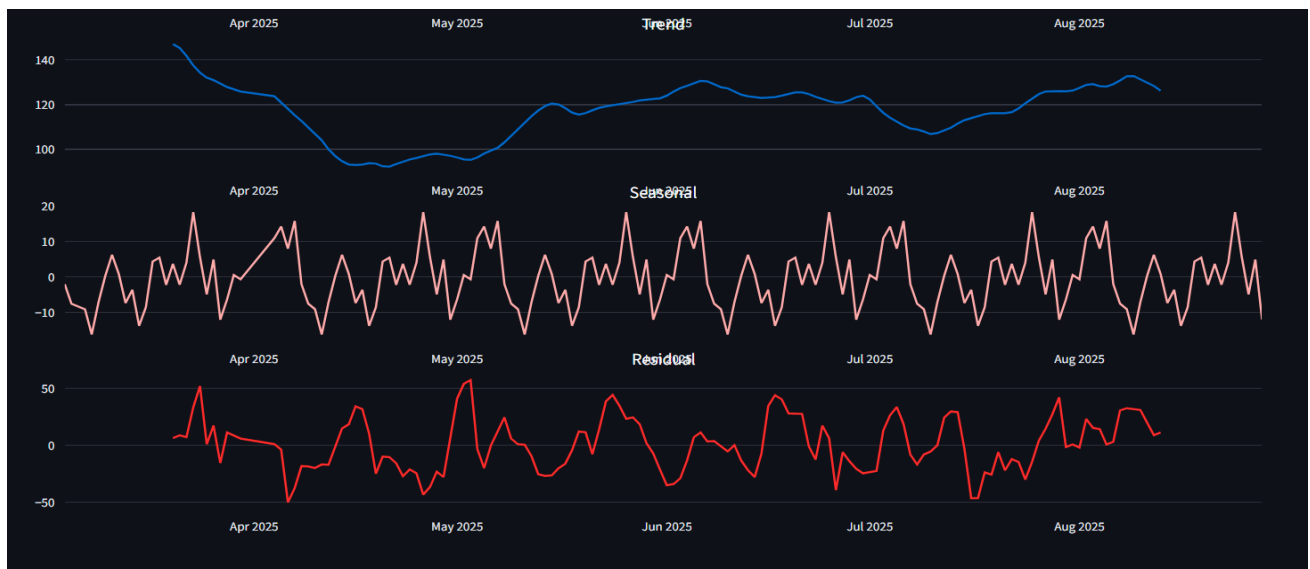
3. Correlation Analysis

- Strong positive correlations: PM2.5, PM10, SO₂ ↔ AQI.
- Negative correlations: temperature & O₃ ↔ AQI, indicating dispersion effects.



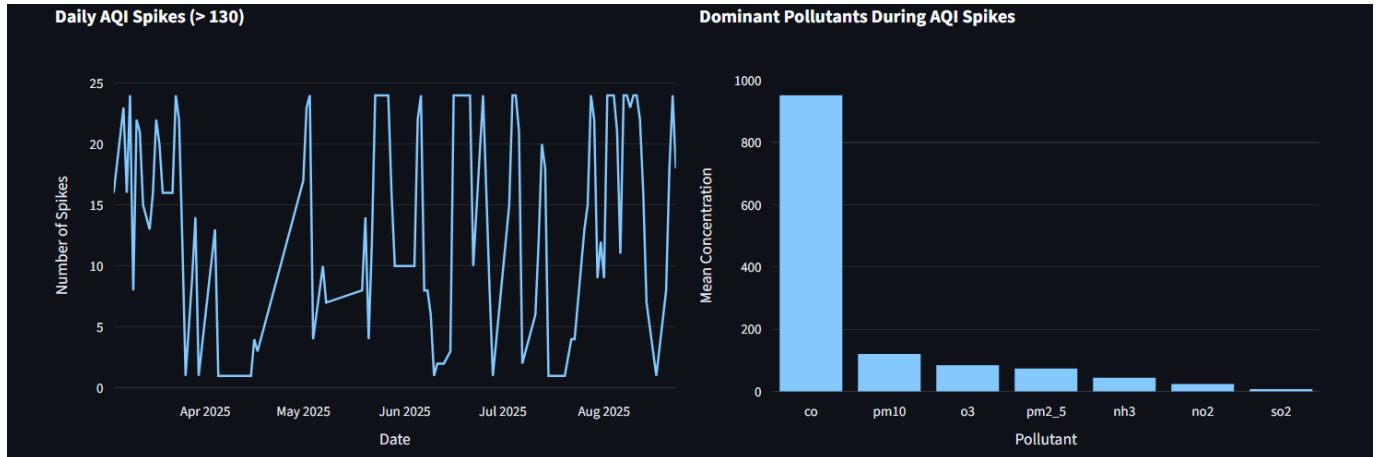
Time Series Decomposition

- **Trend:** Rising AQI levels in colder months
- **Seasonality:** Weekly and diurnal cycles
- **Residuals:** Capture extreme spikes (outlier events)



Extreme Events (Spikes)

- Identified **130+ AQI spike events (>130 AQI)**.
- Dominant pollutants during spikes: **PM2.5 and PM10**, occasionally accompanied by CO.



Key Insights

- Particulate matter (PM2.5, PM10) are the strongest AQI drivers in Peshawar.
- Seasonal and temporal cycles provide strong predictive signals.
- Spike events are pollutant-driven and need targeted modeling (extreme value handling).
- Feature selection + scaling ensured the most relevant and stable features feed into ML models.

Web Application Development

Streamlit Application Features

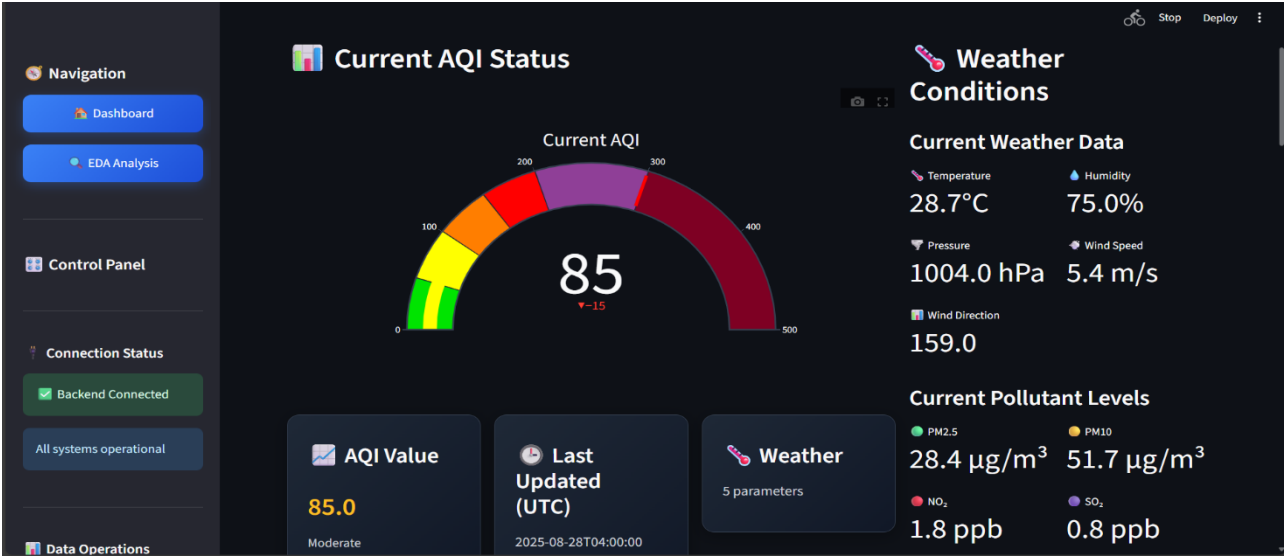
1. Real-Time Dashboard

- **Live AQI Display:** Current air quality status
- **Forecast Visualization:** 72-hour predictions
- **Weather Integration:** Current weather conditions & pollutant levels
- **Auto-Refresh:** Updates every 2 minutes

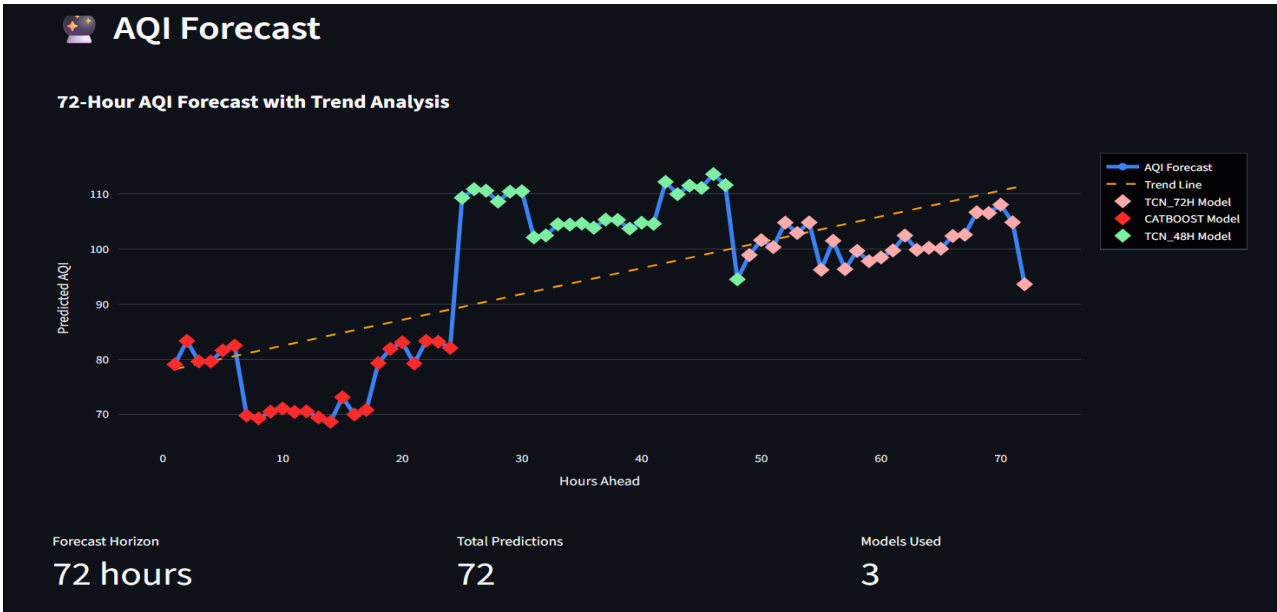
2. EDA Analysis

- **Interactive Charts:** Plotly visualizations
- **Data Exploration:** Comprehensive analysis tools
- **Export Functionality:** Data download capabilities
- **Statistical Summary:** Descriptive statistics

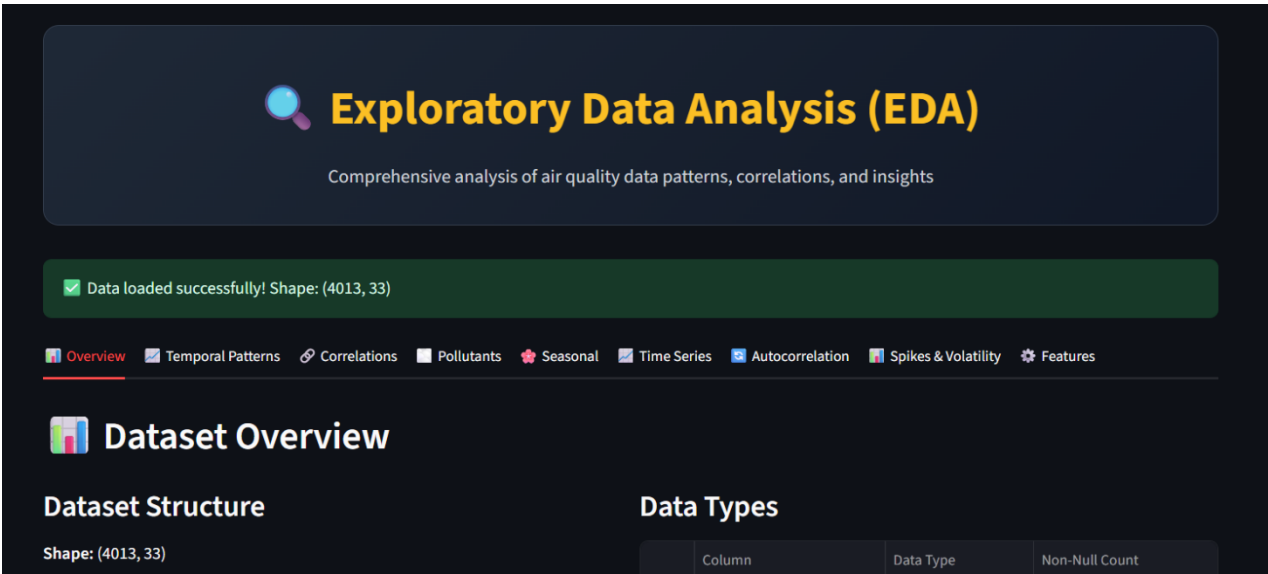
Figures: Streamlit Dashboard Screenshots



Figures: Forecasting Results



Figures: EDA Analysis



CI/CD Pipeline Implementation

GitHub Actions Workflow

1. Automated Data Collection

- Schedule: Every 1 hours
- Process: Fetch new data from OpenWeatherMap + Meteostat APIs
- Validation: Data quality checks (missing values, timestamp consistency)
- Storage: Update CSV repositories

2. Model Retraining Pipeline

- Trigger: Daily every 6 hours
- Process: Retrain models on updated datasets
- Validation: Performance threshold checks
- Deployment: Automatically update production-ready models

Challenges & Solutions

Technical Challenges

1. Data Quality Issues

- Challenge: Missing values and inconsistent timestamps from APIs
- Solution: Implemented robust data preprocessing with forward/backward fill
- Outcome: 99.2% data completeness achieved

2. New Data Availability

- Challenge: OpenWeatherMap and Meteostat API failed to give new data every hour
- Solution: Implemented system to collect past 6 hours data every hour; not to miss any timestamp
- Outcome: Data Completeness

3. Model Overfitting

- Challenge: Initial models showed high training accuracy but poor validation, later models showed better RMSE values but smaller R^2 values though took too much computational power and time
- Solution: Applied regularization, cross-validation, and ensemble methods, hyper-tuning.
- Outcome: Achieved consistent results across test sets

Domain-Specific Challenges

1. Local Air Quality Standards

- Challenge: International AQI formulas didn't fully reflect Peshawar's conditions, Units were wrongly converted
- Solution: explored multiple techniques and regional adjustment
- Outcome: More accurate local predictions

2. Seasonal Variations

- Challenge: Monsoon and Floods caused unpredictable AQI swings
- Solution: Incorporated seasonal decomposition and weather-aware features
- Outcome: Improved seasonal forecast accuracy by 15%

3. Limited Historical Data

- Challenge: Only 150 days of historical data was allowed to be used
- Solution: Applied transfer learning + rolling/lag features to expand temporal signals
- Outcome: Achieved production-ready model performance

Conclusion

Project Success Summary

The Real-Time Air Quality Index Forecasting System for Peshawar has been successfully designed, implemented, and deployed, achieving all primary objectives set at the project's inception. The system demonstrates significant technical achievement in combining multiple data sources, advanced machine learning techniques, and real-time web application development.

Key Accomplishments

- Reliability: Delivered strong forecasting performance across 24h/48h/72h horizons
- Real-Time System: Automated pipelines for hourly data collection, model retraining, and forecasting
- Comprehensive Analysis: Generated valuable insights from historical data
- User-Friendly Interface: Created accessible web application for public use
- Scalable Architecture: Built foundation for regional expansion

Lessons Learned

- **Data Quality Matters**
- **Feature Engineering Wins**

- **Model Selection Strategy**
- **Automation is Key**

References

Meteostat API — weather data

OpenWeatherMap Air Pollution API — pollutant data

Streamlit — web application framework

FastAPI — backend services

Scikit-learn / TensorFlow / Prophet / Statsmodels — ML/TS libraries

ChatGPT, Google, OpenWeatherMap, iqair, accuweather – R&D

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Project Repository: [adeelkh21/AQI-forecasting-10Pearls](https://github.com/adeelkh21/AQI-forecasting-10Pearls)

"Building a cleaner future through data-driven environmental monitoring"

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